# CHEMISTRY

**SECTION - A**

**Multiple Choice Questions:** This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

**Choose the correct answer:**

1. The correct order of ionisation energy of

   Cl, S, P, Al, Si is

   (1) Cl > P > S > Si > Al

   (2) P > Cl > S > Al > Si

   (3) Cl > S > P > Si > Al

   (4) Cl > Al > Si > P > S

**Answer (1)**

**Sol.** **kJ/mol**

Cl ⇒ 1256

S ⇒ 999

P ⇒ 1012

Al ⇒ 577

Si ⇒ 786

2. Consider the statements below :

   **Statement I :** $[CoBr_4]^{2-}$ absorbs lesser energy than $[CoCl_4]^{2-}$.

   **Statement II :** $[CoCl_4]^{2-}$ has higher crystal field splitting energy than $[CoBr_4]^{2-}$.

   (1) Both statement I and statement II are correct

   (2) Both statement I and statement II are incorrect

   (3) Statement I correct and statement II incorrect

   (4) Statement I incorrect and statement II correct

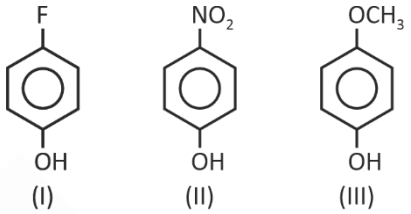
**Answer (1)**

**Sol.** Order of ligand strength $Cl^- > Br^-$

$[CoCl_4]^{2-}$ absorb more energy than $[CoBr_4]^{2-}$

$[CoCl_4]^{2-}$ has more CFSE value due to strength of ligand.

3. Arrange in increasing order of acid strength.



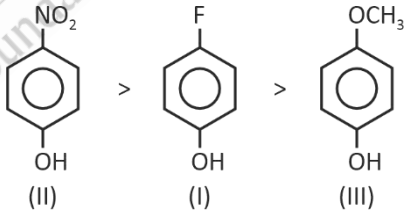

   (1) (II) > (I) = (III)

   (2) (I) > (II) > (III)

   (3) (II) > (I) > (III)

   (4) (II) = (I) > (III)

**Answer (3)**

**Sol.** $K_a$ order 


4. Which of the following undergo nitration at fastest rate?

   (1) $C_6H_5NO_2$

   (2) $C_6H_5CH_3$

   (3) $C_6H_5COOH$

   (4) $C_6H_5Br$

**Answer (2)**

**Sol.** Methyl group is activating group



5. For the following change,

$$H_2O(\ell) \underset{5^\circ C}{\rightarrow} H_2O(g) \quad (100^\circ C)$$

Select the correct answer:

(1) q = +ve, w = +ve, $\Delta H$ = +ve

(2) q = –ve, w = –ve, $\Delta H$ = +ve

(3) q = +ve, w = –ve, $\Delta H$ = +ve

(4) q = –ve, w = –ve, $\Delta H$ = –ve

**Answer (3)**

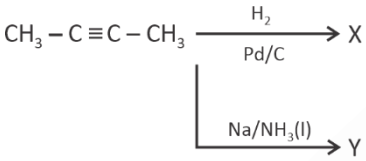
**Sol.** $H_2O(\ell) \underset{5^\circ C}{\rightarrow} H_2O(g) \quad (100^\circ C)$

$\rightarrow \because$ expansion w = –ve.

$\rightarrow$ It is heat absorbing, so, q = +ve.

$\Delta H$ = +ve

6. Consider the following reaction




Choose the correct option.

(1) X $\Rightarrow$ $CH_3–CH_2–CH_2–CH_3$

Y $\Rightarrow$ $CH_3 – CH = CH – CH_3$ (cis)

(2) X $\Rightarrow$ $CH_3 – CH = CH – CH_3$ (cis)

Y $\Rightarrow$ $CH_3 – CH = CH – CH_3$ (trans)

(3) X $\Rightarrow$ $CH_3 – CH = CH – CH_3$ (cis)

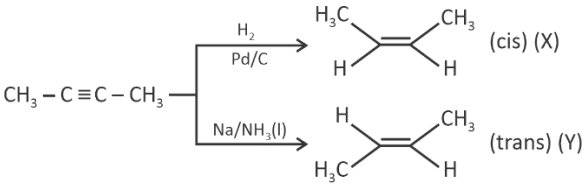
Y $\Rightarrow$ $CH_3 – CH = CH – CH_3$ (cis)

(4) X $\Rightarrow$ $CH_3 – CH = CH – CH_3$ (trans)

Y $\Rightarrow$ $CH_3 – CH = CH – CH_3$ (cis)

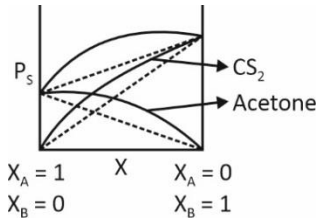
**Answer (2)**

**Sol.**

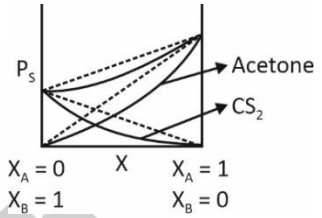



7. A binary solution is formed by mixing Acetone (A) and $CS_2$ (B). The variation of vapour pressure v/s mole fraction will be

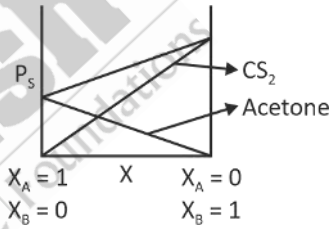
(1)



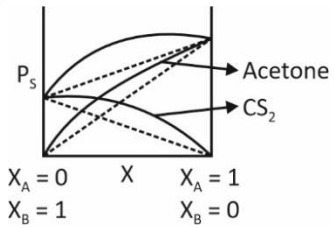

(2)




(3)




(4)




**Answer (1)**

**Sol.** $P^o_{CS_2} \approx 510\,mm\,Hg$

$P^o_{Acetone} \approx 310\,mm\,Hg$

$CS_2$ and Acetone shows positive deviation

$P_S > P^o_A X_A + P^o_B X_B$



8. Match the List-I with List-II and choose the correct option :

| | List-I | | List-II |
|---|---|---|---|
| (a) | 2nd orbit of $He^+$ ion | (i) | $-1.96 \times 10^{-17}$ J |
| (b) | 3rd orbit of H-atom | (ii) | $-2.42 \times 10^{-19}$ J |
| (c) | 1st orbit of $Li^{2+}$ ion | (iii) | $-2.178 \times 10^{-18}$ J |
| (d) | 2nd orbit of $Li^{2+}$ ion | (iv) | $-4.9 \times 10^{-18}$ J |

(1) (a)-(iii); (b)-(ii); (c)-(i); (d)-(iv)

(2) (a)-(iii); (b)-(ii); (c)-(iv); (d)-(i)

(3) (a)-(iv); (b)-(iii); (c)-(ii); (d)-(i)

(4) (a)-(i); (b)-(ii); (c)-(iii); (d)-(iv)

**Answer (1)**

**Sol.** $E_n = -2.178 \times 10^{-18} \frac{z^2}{n^2} J$

(a) z – 2, n = 2

(b) z = 1, n = 3

(c) z = 3, n = 1

(d) z = 3, n = 2

9. Given below are two statements.

Statement I: Sublimation is a purification technique that is used to separate those solid substances which changes from solid to vapour state without passing through liquid state.

Statement II: If external atmospheric pressure is reduced, then boiling point of substance is decreased.

In the light of the above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

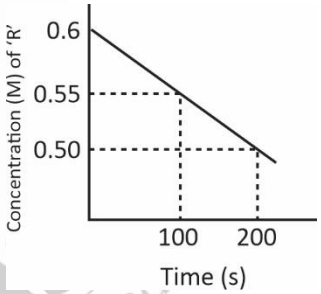
(4) Statement I is incorrect but statement II is correct

**Answer (1)**

**Sol.** Sublimable compounds are those compounds which changes from solid to vapour state without passing through liquid state.

Boiling point depends on external atmospheric pressure.

10. Consider the following graph of concentration vs time, of a reaction, R → P




Find half-life of reaction.

(1) 600 s

(2) 200 s

(3) 300 s

(4) 100 s

**Answer (1)**

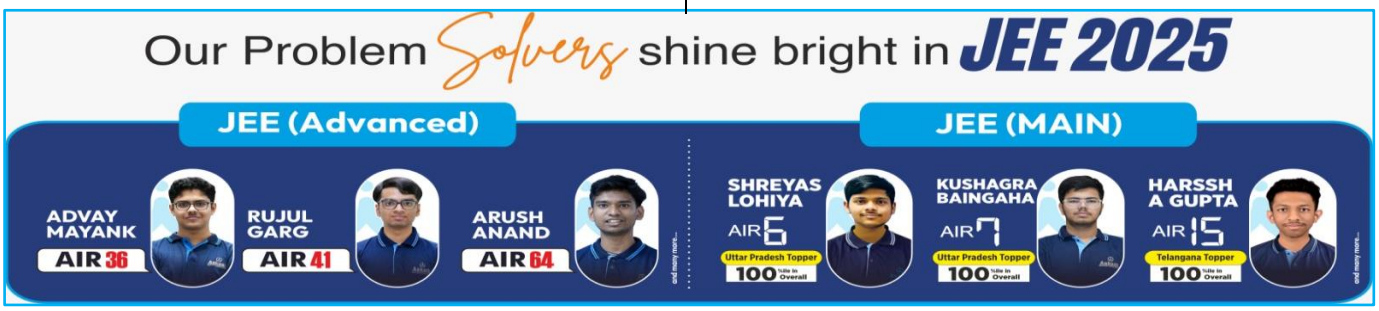
**Sol.** The graph shows straight line for concentration vs time.

For zero order

$A_t = -kt + A_0$

At, t = 100 s, conc. = 0.55

At t = 200 s, conc. = 0.5



$$k = -\frac{\text{Change in concentration}}{\text{Change in time}}$$

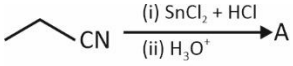
$$= \frac{0.5-0.55}{200-100} = \frac{0.05}{100}$$

$k = 5 \times 10^{-4}\ ms^{-1}$

$$t_{\frac{1}{2}} = \frac{A_0}{2k}$$

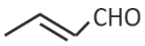
$$= \frac{0.6}{2\times5\times10^{-4}}$$

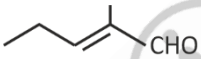
$$= \frac{6000}{100} = 600\ s$$

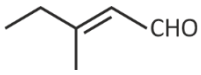


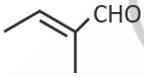

11. $CH_3 - CH = CH - CH_3 \xrightarrow[(ii)\ Zn\text{-}H_2O]{(i)\ O_3} A$

A and B are mixed and treated with dil. base to give mixture of products. Choose the incorrect product.

(1) 


(2) 


(3) 


(4) 


**Answer (3)**

**Sol.** $CH_3 - CH_2 - CN \xrightarrow[(ii)\ H_3O^+]{(i)\ SnCl_2+HCl} \underset{(A)}{CH_3 - CH_2 - CHO}$

$$CH_3 - CH = CH - CH_3 \xrightarrow[Zn\text{-}H_2O]{O_3} \underset{(B)}{CH_3 - CHO}$$

$$CH_3 - CH_2 - CHO + CH_3 - CHO \xrightarrow{OH^-}$$

$CH_3 - CH_2 - CH = CH - CHO$

+

$CH_3 - CH = CH - CHO$

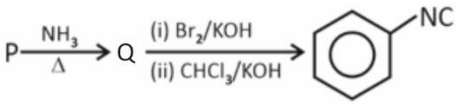
+

$CH_3 - CH(CHO) = CH - CH_3$

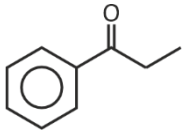
+

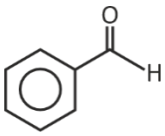
$CH_3 - CH(CHO) = CH - CH_2 - CH_3$

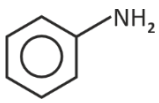
12. Consider the following reaction :

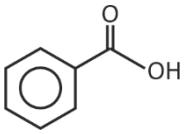



The structure of 'P' is

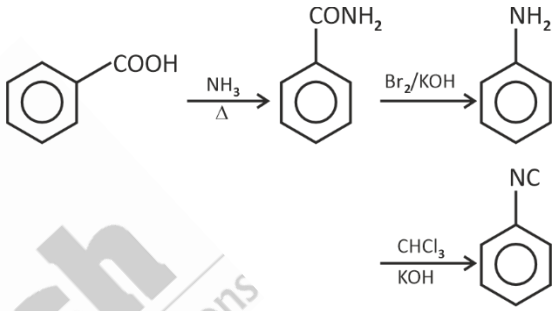
(1) 


(2) 


(3) 


(4) 


**Answer (4)**

**Sol.** 


13. Consider the two complexes

I : $[Cr(OX)_3]^{3-}$

II : $[Co(OX)_3]^{3-}$

Find the ratio of CFSE of I to II complex (neglect pairing energy and consider $\Delta_o$ for both complexes to be x)

(1) 2 (2) $\frac{1}{2}$

(3) $\frac{1}{3}$ (4) $\frac{2}{3}$

**Answer (2)**

**Sol.** $\left[Cr(OX)_3\right]^{3-} \Rightarrow Cr^{3+} \Rightarrow 3d^3 = t_{2g}^3 e_g^0$

CFSE = 3 × (–0.4) $\Delta_o$ = –1.2$\Delta_o$

$\left[Co(OX)_3\right]^{3-} \Rightarrow Co^{3+} \Rightarrow 3d^6 \Rightarrow t_{2g}^6 e_g^0$



CFSE = $-0.4 \times 6\ \Delta_o = -2.4\ \Delta_0$

Ratio of CFSE $= \frac{-1.2\Delta_o}{-2.4\Delta_o} = \frac{1}{2}$

14. Select the correct statement about 13th group elements.

A. Electronegativity decreases regularly down the graph.

B. Ionic radii decreases down the graph.

C. Boron has highest ionisation energy.

D. Trichloride of aluminium are covalent in nature.

(1) A, B, C only (2) C, D only

(3) A, C, D only (4) B, C, D only

**Answer (2)**

**Sol.** (A) is false

EN → B > Al < Ga < In < Tl

2 1.5 1.6 1.7 1.8

(B) is false

Ionic radii → $B^{3+} < Al^{3+} < Ga^{3+} < In^{3+} < Tl^{3+}$

(C) is true

| I.E. → | B | Al | Ga | In | Tl |
|---|---|---|---|---|---|
| ($kJ/mol^{-1}$) | 800 | 577 | 579 | 558 | 589 |

15. Match List-I with List-II.

| | List-I | | List-II |
|---|---|---|---|
| A. | Bayer's unsaturation test | (I) | Violet/purple colour |
| B. | Cerric ammonium nitrate test of alcohols | (II) | Red colour |
| C. | Tollen's reagent test | (III) | Silver mirror obtained |
| | $FeCl_3$ test of phenol | (IV) | Pink colour discharge |

Select the correct option.

(1) A(II), B(I), C(IV), D(III)

(2) A(II), B(I), C(III), D(IV)

(3) A(IV), B(II), C(III), D(I)

(4) A(IV), B(III), C(II), D(I)

**Answer (3)**

**Sol.** A. Bayer's unsaturation test → Pink colour discharges

B. Cerric ammonium nitrate test of alcohols → The colour of solution changes from yellow to red

C. Tollen's reagent test → Silver mirror (Ag↓) is observed

D. $FeCl_3$ test of phenol → Violet colour complex is formed

16. Consider the given cell

$Ag(s)|AgCl(s)|Cl^-(aq)|FeCl_2(aq), FeCl_3(aq)|Pt(s)$

In which of the following cases, $E_{cell}$ will increase

(1) $[Fe^{2+}]$ increases
$[Cl^-]$ increases

(2) $[Fe^{2+}]$ increases
$[Cl^-]$ decreases

(3) $[Fe^{3+}]$ & $[Cl^-]$ increase

(4) $[Fe^{2+}]$ decreases
$[Fe^{3+}]$ decreases

**Answer (3)**

**Sol.**

Anode $Ag(s) + Cl^-_{(aq)} \rightarrow AgCl(s) + e^-$

Cathode $Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$

$$Ag(s) + Cl^-(aq) + Fe^{3+}(aq) \rightarrow Fe^{2+}(aq) + AgCl(s)$$

$$Q = \frac{[Fe^{2+}]}{[Fe^{3+}][Cl^-]}$$



$$E_{cell} = E^\circ_{cell} - \frac{0.06}{1}\log\frac{[Fe^{2+}]}{[Fe^{3+}][Cl^-]}$$

$E_{cell}$ increase if $[Fe^{3+}]$ & $[Cl^-]$ increases

$E_{cell}$ decreases if $[Fe^{2+}]$ increases

17. Consider the following statements and choose the correct option(s)

(A) $Ni^{2+}$ forms cherry red colour complex with dimethyl glyoximate ligand (dmg)

(B) $Ni^{2+}$ complex with dmg contains two ring of five membered

(C) $Ni^{2+}$ has 2 unpaired electron in d-orbitals in the complex

(D) $Ni^{2+}$ complex with dmg is soluble at $p^H = 9$

(1) Only A & B

(2) Only B & C

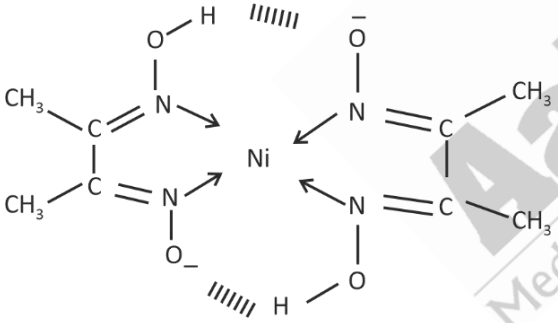
(3) Only A & C

(4) Only A, C and D

**Answer (1)**

**Sol.** $Ni^{2+} + 2dmg \rightarrow [Ni(dmg)_2]$

1. $[Ni(dmg)_2]$ is of pink colour



2. 2 five membered ring is present.

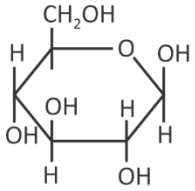
3. $Ni^{2+}$ with dmg for $dsp^2$ hybridised square planar complex

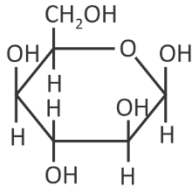
$Ni^{2+} \Rightarrow 3d^8$

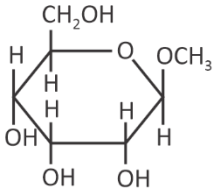
no unpaired electron

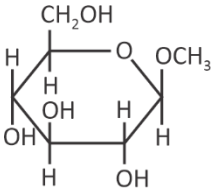
4. $[Ni(dmg)_2]$ is soluble at low pH

18. Consider the following molecules.

(A) 

(B) 

(C) 

(D) 

The examples of non-reducing sugar(s) are

(1) A, B only (2) A, C only

(3) B, D only (4) C, D only

**Answer (4)**

**Sol.** (A), (B) are reducing sugars as it contains hemiacetal linkage.

(C), (D) are non-reducing sugar as it contain acetal linkage.

19. Following molecules are given :

$HNO_3$, $NF_3$, $H_2SO_4$, $O_3$

Consider the molecule (T) having maximum number of lone pairs (on all atoms).

The bond angle (∠XMX), where M is central atom in T is

(1) 110° (2) 97°

(3) 102° (4) 115°

**Answer (3)**

**Sol.** $NF_3$ has 10 lone pairs.

Bond angle is 102.3°.

20.

**SECTION - B**

**Numerical Value Type Questions:** This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.



21. In the reaction sequence, what is the mass (in grams) of product (C) formed?

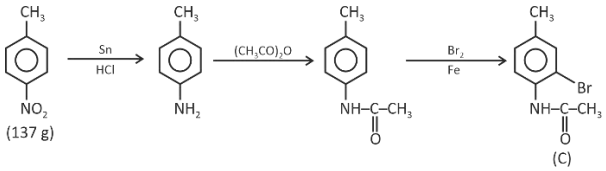



(A)

(137 g)

**Answer (228)**

**Sol.**




moles of (A) = $\frac{137}{137} = 1$ mole.

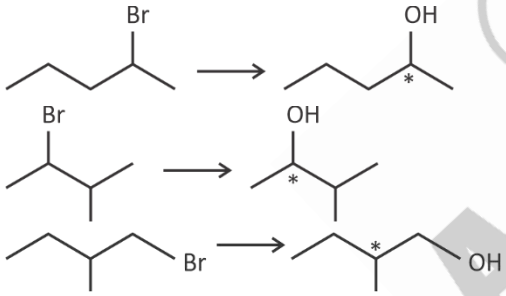
moles of (C) formed = 1 moles

mass of (C) = 1 × 228 = 228 (g)

22. $C_5H_{11}Br$ reacts with aq. KOH without rearrangement. How many optically active compounds are formed.

**Answer (3)**

**Sol.**




23. x g of pure $Cl_2$ is reacted with $Ba(OH)_2$ to form $Ba(ClO_3)_2$. $Ba(OH)_2$ concentration is 1 M and volume is 25 mL. Find x.

**Answer (2)**

**Sol.** $6Cl_2 + 6Ba(OH)_2 \rightarrow Ba(ClO_3)_2 + 5BaCl_2 + 6H_2O$

1 M × 25 mL

= 25 m mol

6 mol of $Ba(OH)_2$ reacted with 6 mol $Cl_2$

25 m mol of $Ba(OH)_2$ reacted with 25 m mol of $Cl_2$

mol of $Cl_2$ reacted = $25 \times 10^{-3}$ mol

mass of $Cl_2 = 25 \times 10^{-3} \times 71 = 1.775\ g \approx 2g$

24. For the reaction,

$$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$$

Total pressure at equilibrium = $\sqrt{3}$ atm

$K_p$ = 9 atm

$\alpha = x \times 10^{-1}$

Use $\sqrt{2} = 1.41$

Find x. (Report to nearest integer)

**Answer (9)**

**Sol.**

$$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$$

$t = 0$ P

$t = t_{eq}$ $P - \alpha P$ $\frac{\alpha P}{2}$ $\frac{3\alpha P}{2}$

$P + \alpha P = \sqrt{3}$ ...(1)

$$9 = K_p = \frac{\left(\frac{3\alpha P}{2}\right)^{3/2}\left(\frac{\alpha P}{2}\right)^{1/2}}{P(1-\alpha)} \quad ...(2)$$

$$P(1+\alpha) = \sqrt{3} \Rightarrow P = \frac{\sqrt{3}}{1+\alpha}$$

$$\Rightarrow 9 = \frac{\left[\frac{3\alpha \times \sqrt{3}}{2(1+\alpha)}\right]^{3/2}\left[\frac{\alpha \times \sqrt{3}}{1+\alpha}\right]^{1/2}}{\frac{\sqrt{3}}{1+\alpha}[1-\alpha]}$$

$$= \frac{\left[\frac{3\sqrt{3}}{2}\right]^{3/2}[\sqrt{3}]^{1/2}\left[\frac{\alpha}{1+\alpha}\right]^{3/2}\left[\frac{\alpha}{1+\alpha}\right]^{1/2}}{(\sqrt{3})\left(\frac{1-\alpha}{1+\alpha}\right)}$$

Solving $\alpha$ = 0.86

= $8.6 \times 10^{-1}$

25.



# MATHEMATICS

## SECTION - A

**Multiple Choice Questions:** This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

**Choose the correct answer :**

1. A rectangle is formed by lines $x = 0$, $y = 0$, $x = 3$, $y = 4$. A line perpendicular to $3x + 4y + 6 = 0$ divides the rectangle into two equal parts, then the distance of the line from $\left(-1, \frac{3}{2}\right)$ is

(1) 2 (2) $\frac{17}{10}$

(3) $\frac{6}{5}$ (4) $\frac{8}{5}$

**Answer (2)**

**Sol.** Concept : line passes through centre of the rectangle $\equiv \left(\frac{3}{2}, 2\right)$

$\Rightarrow 4x - 3y = C$ passes through $\left(\frac{3}{2}, 2\right)$

$\Rightarrow C = 0$

$\Rightarrow$ distance of $\left(-1, \frac{3}{2}\right)$ from $4x - 3y = 0$ is $\frac{\left|-4 - \frac{9}{2}\right|}{5} = \frac{17}{10}$

2. Let $A = \{-2, -1, 0, 1, 2, 3, 4\}$ and $R$ be a relation $R$, such that $R = \{(x, y) : (2x + y) \le -2, x \in A, y \in A\}$

Let $l$ = number of elements in R

$m$ = minimum number of elements to be added in R to make it reflexive

$n$ = minimum number of elements to be added in $R$ to make it symmetric, then $(l + m + n)$ is

(1) 10 (2) 17

(3) 11 (4) 14

**Answer (2)**

**Sol.** $R = \{(x, y) : 2x + y \le -2, x \in A, y \in A\}$

$A = \{-2, -1, 0, 1, 2, 3, 4\}$

$R = \{(-2, 2), (-2, 1), (-2, 0), (-2, -1), (-2, -2), (-1, 0), (-1, -1), (-1, -2), (0, -2)\}$

$l = 9$

$m = 5$

$n = \{(-2, 1), (-2, 2), (-1, 0)\} = 3$

3. Number of 4 letter words with or without meaning formed from the letters of the word PQRSSSTTUVV is

(1) 1232 (2) 1400

(3) 1422 (4) 1162

**Answer (3)**

**Sol.** Case A : The word is ABCD

$\Rightarrow {}^7C_4 \cdot 4! = 840$

Case B: The word is AAAB

$\Rightarrow (SSS) \times ({}^6C_1) \Rightarrow {}^6C_1 \cdot \frac{4!}{3!} = 24$

Case C : The word is AABB

$\Rightarrow {}^3C_2 \cdot \frac{4!}{2!2!} = 18$

Case (D) : The word is AABC

$\Rightarrow \left({}^3C_1\right) \cdot \left({}^6C_2\right) \cdot \frac{4!}{2!} = 3 \times 15 \times 12 = 540$

Answer 1422

4. The area (in square units) bounded by the curve $f(x) = \max\{\sin x, \cos x\}$ and x-axis between $x = 0$ and $x = \frac{3\pi}{2}$ is $A$ sq. unit. Then, the value of $A + A^2$ is

(1) $8 - 2\sqrt{2}$ (2) $8 + 2\sqrt{2}$

(3) $6 - 2\sqrt{2}$ (4) $6 + 2\sqrt{2}$

**Answer (1)**

